AI Milk Frog Companion

A friendly physical companion for everyone

Voice companionship and gentle support through study, cooking and practical tasks.



PROJECT PROPOSAL

Daily companionship for everyone

For people who study, cook or follow practical workflows

01

Studying

Studying alone can make keeping a routine harder.

02

Cooking

Hands can be busy while preparing a meal.

03

Lab practice

Practical tasks require following unfamiliar steps.

For everyone, with modes chosen for each person's routine.
One familiar character offers companionship throughout the day.

One companion for everyone with three modes

STUDY

Follow the plan

Desk-presence cues
Scheduled breaks
Encouraging reminders

COOKING

Follow the recipe

Ingredient timing
Step confirmation
Multiple timers

EXPERIMENT

Follow the protocol

Approved checklists
Step confirmations
Supervisor support

Across all modes: brief voice conversations, gentle encouragement and adjustable reminders.

Study Companion

A familiar voice through study sessions

“Study calculus for 25 minutes,
then take a 5-minute break.”

Notice sustained absence from the desk.
Ask whether to pause or continue.
Offer encouragement, snooze or quiet mode.



Cooking Assistant



Cooking support for everyday life

Choose a reviewed recipe.
Use a kitchen where cooking is permitted.

Speak the next approved step.
Track timers and confirm by voice.
Offer friendly help while hands are busy.

Experiment Assistant

A practical companion for work and learning

Use an approved checklist
and obtain permission to use the device.

Read approved steps and confirm by voice.
Use visible cues only as supporting hints.
Ask a supervisor about uncertain steps.



Competition direction

Everyday AI

Battlefield Lightning

ASUS UGen300 AI Accelerator

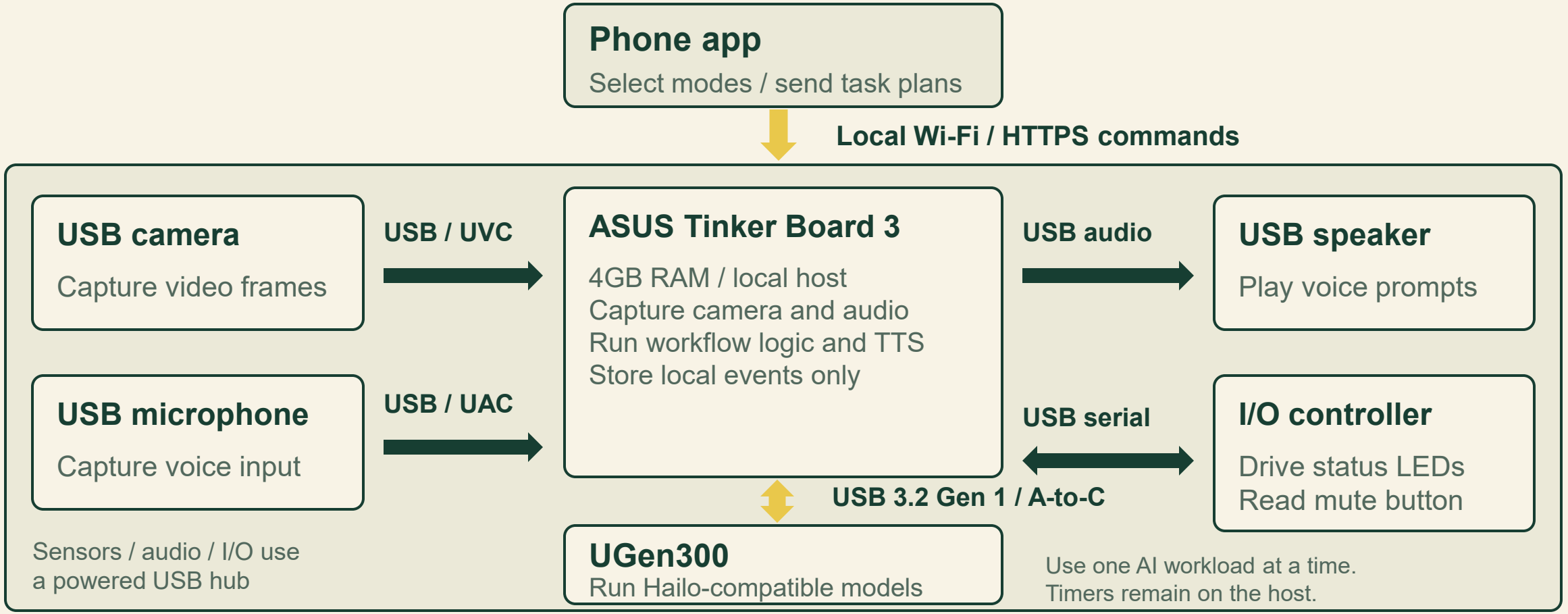
Local perception and voice dialogue support everyday routines.
The host controls approved workflows, timers and reminders.

App connection and feature activation

01	**Pair**	Pair over local Wi-Fi or a phone hotspot using a device code.
02	**Choose**	Use included study and voice features, or buy optional packs.
03	**Plan**	Select study, cooking or an approved lab workflow.
04	**Assist**	Talk to Milk Frog and receive local task reminders.
05	**Manage**	Pause or switch modes. The device speaks the session summary.

Hardware architecture

Desktop architecture with a ventilated compute base



Local only: camera, audio, AI results and session records never reach the app or internet.

Software architecture

App commands enter. Monitoring data stays local.

Flutter app + account service

Feature purchases and ownership

Verified feature + task command

No camera, audio, or monitoring data returns

LOCAL PROCESSING PIPELINE



Capture

Camera + audio

Interpret

YOLO + Whisper
Qwen on demand

Track

Approved JSON
workflows + timers

Decide

User confirmation
+ reminder cooldown

Respond

Piper speech
+ local event records

Camera, audio, AI results, and session records never leave the local host.

Selected AI models

Role	Selected model	On-device use
Vision	YOLOv8n	UGen300: person and common object cues
Speech recognition	Whisper Base	UGen300: spoken commands and step confirmations
Companionship	Qwen2.5-1.5B-Instruct	UGen300: brief dialogue and encouragement
Scene questions	Qwen2-VL-2B-Instruct	UGen300: on-demand image questions with confirmation
Voice output	Piper TTS + owned recordings	Host CPU: local Mandarin speech and fixed reminder audio

Hailo-10H model builds. Load AI tasks sequentially to limit memory use.

Software and model coordination

Layer	Selected tools	Responsibility
Capture / AI	OpenCV + HailoRT USB	Camera/audio input; Hailo model execution
Model manager	Python task queue	Run one AI workload at a time; resume vision
Task engine	Python transitions + JSON	Approved steps, timers and reminder cooldown
Local storage	SQLite + owned audio files	Local events and preferences; user deletion
Phone app	Flutter + local HTTPS API	Pairing, mode selection and pack purchases
Account service	FastAPI + Pydantic	Receipt checks; signed offline entitlements

Timers and fixed alerts continue during model switching. No monitoring data reaches the account service.

Privacy and reliability

Monitoring data stays on the device

Sensing and dialogue remain local.
Physical shutter, mute and quiet mode.
No raw recording by default.

01 Data

Camera, audio, and monitoring data stay on the device.

02 Uncertainty

Ask for confirmation when a scene or step is unclear.

03 Failure

Local timers continue without the app or internet.

04 Safety

Approved instructions and instructor oversight for labs.

Business model

For everyone + optional packs

Everyday companionship for everyone

Study and voice companionship are included.
Add recipe or course packs when needed.
One device adapts to each person's needs.

One-time pack purchases keep costs predictable.
Local AI has no per-conversation cloud charge.

Licensed IP collaborations

Character editions can broaden product appeal

Start with an owned or authorized Milk Frog design.
Later add licensed shells and themed content.
Secure permissions before each collaboration.

Develop an affordable basic edition for everyone.
Optional editions add value to the same platform.
Final pricing depends on production cost.

Expected outcomes

A familiar companion through everyday routines

Expected benefits

Friendly voice companionship.
Support for study and daily independence.
Easier access to approved next steps.

Proposed system scope

Milk Frog desktop device and companion app.
Shared voice engine with three task modes.
Local workflow files and source repository.

Future features for everyday life

A feature library that grows with everyday needs

Daily planning

Study or work blocks and task reminders

Practical skills

Approved assembly steps and tool checklists

Home plant care

Care schedules and visible leaf-change prompts

Language practice

Role-play conversations and speaking prompts

Habit support

Hydration and routine reminders

Exercise routines

General movement timing and rest prompts